A COHOMOLOGICAL VANISHING HYPOTHESIS ON A NOETHERIAN ALGEBRAIC SPACE FORCES IT TO BE A SCHEME AND THE INVERTIBLE SHEAF TO BE AMPLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- A space is affine as soon as the first cohomology of every quasi-coherent module vanishes. Same chapter spaces-cohomology.tex, label proposition-vanishing-affine (invoked at proof line 3248). It reduces affineness of the open U below a section to the vanishing $H^1(U, \mathcal{F}) = 0$ for all quasi-coherent $\mathcal{F}$.
- The inclusion of the nonvanishing open of a section is affine. Morphisms, label morphisms-lemma-affine-s-open (invoked at proof line 3251) and Morphisms of Spaces, label spaces-morphisms-lemma-affine-local (invoked at proof line 3252). They show that $j : U \to X$ is affine, which is what turns $H^1(U, -)$ into $H^1(X, j_*-)$.
- Vanishing of the higher direct images of an affine morphism, and the Leray identification. Same chapter spaces-cohomology.tex, label lemma-affine-vanishing-higher-direct-images (invoked at proof line 3258) and Cohomology on Sites, label sites-cohomology-lemma-apply-Leray (invoked at proof line 3259). Together they give $H^1(U, \mathcal{F}) = H^1(X, j_*\mathcal{F})$.
- A quasi-coherent module as a filtered colimit of coherent modules, and cohomology of quasi-compact quasi-separated spaces commuting with such colimits. Same chapter spaces-cohomology.tex, labels lemma-directed-colimit-coherent (invoked at proof line 3262) and lemma-colimits (invoked at proof line 3266). They replace $j_*\mathcal{F}$ by the filtered colimit of coherent $\mathcal{F}_i$ and compute $H^1(X, j_*\mathcal{F})$ as the colimit of the $H^1(X, \mathcal{F}_i)$.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Closed points of a decent Noetherian space, the schematic locus, and Noetherian sobriety. Decent Spaces, label decent-spaces-lemma-decent-space-closed-point (invoked at proof line 3300), Properties of Spaces, label spaces-properties-lemma-subscheme (invoked at proof line 3320) and spaces-properties-lemma-Noetherian-sober (invoked at proof line 3328). They let every closed point be cut out by a closed immersion from a field, show that the closed points lie in the schematic locus once the U are affine, and conclude that an open subset containing all closed points is everything.
- The scheme case of the criterion. Cohomology of Schemes, label coherent-lemma-quasi-compact-h1-zero-invertible (invoked at proof line 3331). Once X is known to be a scheme, it converts the assumed vanishing into ampleness of $\mathcal{L}$.

Source reference. Original lemma 0B5P: sources/stacks-chow/tags.txt, source lines 14042–14042. Navigation pointer for \ref{coherent-lemma-quasi-compact-h1-zero-invertible} used inside the counted span (line 3331); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AHB: sources/stacks-chow/tags.txt, source lines 13224–13224. Navigation pointer for \ref{decent-spaces-lemma-decent-space-closed-point} used inside the counted span (line 3300); the argument that consumes it is retained in the span.

Source reference. Original section 03I7: sources/stacks-chow/tags.txt, source lines 4593–4593. Navigation pointer for \ref{decent-spaces-section-reasonable-decent} used inside the counted span (line 3304); the argument that consumes it is retained in the span.

Source reference. Original lemma 073H: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1522–1522. Navigation pointer for \ref{lemma-affine-vanishing-higher-direct-images} used inside the counted span (line 3258); the argument that consumes it is retained in the span.

Source reference. Original lemma 073E: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 667–667. Navigation pointer for \ref{lemma-colimits} used inside the counted span (line 3266); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UV: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2866–2866. Navigation pointer for \ref{lemma-directed-colimit-coherent} used inside the counted span (line 3262); the argument that consumes it is retained in the span.

Source reference. Original lemma 01SF: sources/stacks-chow/tags.txt, source lines 2336–2336. Navigation pointer for \ref{morphisms-lemma-affine-s-open} used inside the counted span (line 3251); the argument that consumes it is retained in the span.

Source reference. Original proposition 07V6: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3160–3160. Navigation pointer for \ref{proposition-vanishing-affine} used inside the counted span (line 3248); the argument that consumes it is retained in the span.

Source reference. Original lemma 0733: sources/stacks-chow/tags.txt, source lines 9047–9047. Navigation pointer for \ref{sites-cohomology-lemma-apply-Leray} used inside the counted span (line 3259); the argument that consumes it is retained in the span.

Source reference. Original lemma 03WG: sources/stacks-chow/tags.txt, source lines 5111–5111. Navigation pointer for \ref{spaces-morphisms-lemma-affine-local} used inside the counted span (line 3253); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZG: sources/stacks-chow/tags.txt, source lines 6464–6464. Navigation pointer for \ref{spaces-properties-lemma-Noetherian-sober} used inside the counted span (line 3328); the argument that consumes it is retained in the span.

Source reference. Original lemma 03JH: sources/stacks-chow/tags.txt, source lines 4639–4639. Navigation pointer for \ref{spaces-properties-lemma-subscheme} used inside the counted span (line 3320); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume that for every coherent $\mathcal{O}_X$-module $\mathcal{F}$ there exists an $n \geq 1$ such that $H^1(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}) = 0$. Then X is a scheme and $\mathcal{L}$ is ample on X.*

Proof. Let $s \in H^0(X, \mathcal{L}^{\otimes d})$ be a global section. Let $U \subset X$ be the open subspace over which s is a generator of $\mathcal{L}^{\otimes d}$. In particular we have $\mathcal{L}^{\otimes d}|_U \cong \mathcal{O}_U$. We claim that U is affine.

Proof of the claim. We will show that $H^1(U, \mathcal{F}) = 0$ for every quasi-coherent $\mathcal{O}_U$-module $\mathcal{F}$. This will prove the claim by Proposition 07V6. Denote $j : U \to X$ the inclusion morphism. Since étale locally the morphism j is affine (by Morphisms, Lemma 01SF) we see that j is affine (Morphisms of Spaces, Lemma 03WG). Hence we have

$$H^1(U, \mathcal{F}) = H^1(X, j_*\mathcal{F})$$

by Lemma 073H (and Cohomology on Sites, Lemma 0733). Write $j_*\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ as a filtered colimit of coherent $\mathcal{O}_X$-modules, see Lemma 07UV. Then

$$H^1(X, j_*\mathcal{F}) = \operatorname{colim} H^1(X, \mathcal{F}_i)$$

by Lemma 073E. Thus it suffices to show that $H^1(X, \mathcal{F}_i)$ maps to zero in $H^1(U, j^*\mathcal{F}_i)$. By assumption there exists an $n \geq 1$ such that

$$H^1(X, \mathcal{F}_i \otimes_{\mathcal{O}_X} (\mathcal{O}_X \oplus \mathcal{L} \oplus \ldots \oplus \mathcal{L}^{\otimes d-1}) \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}) = 0$$

Hence there exists an $a \geq 0$ such that $H^1(X, \mathcal{F}_i \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes ad}) = 0$. On the other hand, the map

$$s^a : \mathcal{F}_i \longrightarrow \mathcal{F}_i \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes ad}$$

is an isomorphism after restriction to U. Contemplating the commutative diagram

$$\begin{array}{ccc} H^1(X, \mathcal{F}_i) & \longrightarrow & H^1(U, j^*\mathcal{F}_i) \\ \downarrow{\scriptstyle s^a} & & \downarrow{\scriptstyle \cong} \\ H^1(X, \mathcal{F}_i \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes ad}) & \longrightarrow & H^1(U, j^*(\mathcal{F}_i \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes ad})) \end{array}$$

we conclude that the map $H^1(X, \mathcal{F}_i) \to H^1(U, j^*\mathcal{F}_i)$ is zero and the claim holds.

Let $x \in |X|$ be a closed point. By Decent Spaces, Lemma 0AHB we can represent x by a closed immersion $i : \operatorname{Spec}(k) \to X$ (this also uses that a quasi-separated algebraic space is decent, see Decent Spaces, Section 03I7). Thus $\mathcal{O}_X \to i_*\mathcal{O}_{\operatorname{Spec}(k)}$ is surjective. Let $\mathcal{I} \subset \mathcal{O}_X$ be the kernel and choose $d \geq 1$ such that $H^1(X, \mathcal{I} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}) = 0$. Then

$$H^0(X, \mathcal{L}^{\otimes d}) \to H^0(X, i_*\mathcal{O}_{\operatorname{Spec}(k)} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}) = H^0(\operatorname{Spec}(k), i^*\mathcal{L}^{\otimes d}) \cong k$$

is surjective by the long exact cohomology sequence. Hence there exists an $s \in H^0(X, \mathcal{L}^{\otimes d})$ such that $x \in U$ where U is the open subspace corresponding to s as above. Thus x is in the schematic locus (see Properties of Spaces, Lemma 03JH) of X by our claim.

To conclude that X is a scheme, it suffices to show that any open subset of $|X|$ which contains all the closed points is equal to $|X|$. This follows from the fact that $|X|$ is a Noetherian topological space, see Properties of Spaces, Lemma 04ZG. Finally, if X is a scheme, then we can apply Cohomology of Schemes, Lemma 0B5P to conclude that $\mathcal{L}$ is ample. $\square$